

Design and Implement an AI-Powered System
that leverages NLP to detect Adverse Drug
Events(ADE)

AADITH P

Introduction

- Vaccine pharmacovigilance relies on manual review of millions of free-text adverse event reports — a slow, error-prone process.
- Natural Language Processing can automate extraction of Adverse Drug Events (ADEs) from clinical narratives at scale.
- VAERS contains 200K+ reports annually a rich but largely unstructured data source requiring intelligent NLP-based processing.

Problem Statement

- Free-text VAERS reports contain critical safety signals buried in unstructured narratives existing systems miss subtle ADE mentions.
- Manual severity triage of adverse events is subjective, inconsistent, and does not scale to large report volumes.
- No existing pipeline jointly performs ADE/Drug extraction via NER and hierarchical severity classification in an end-to-end manner.

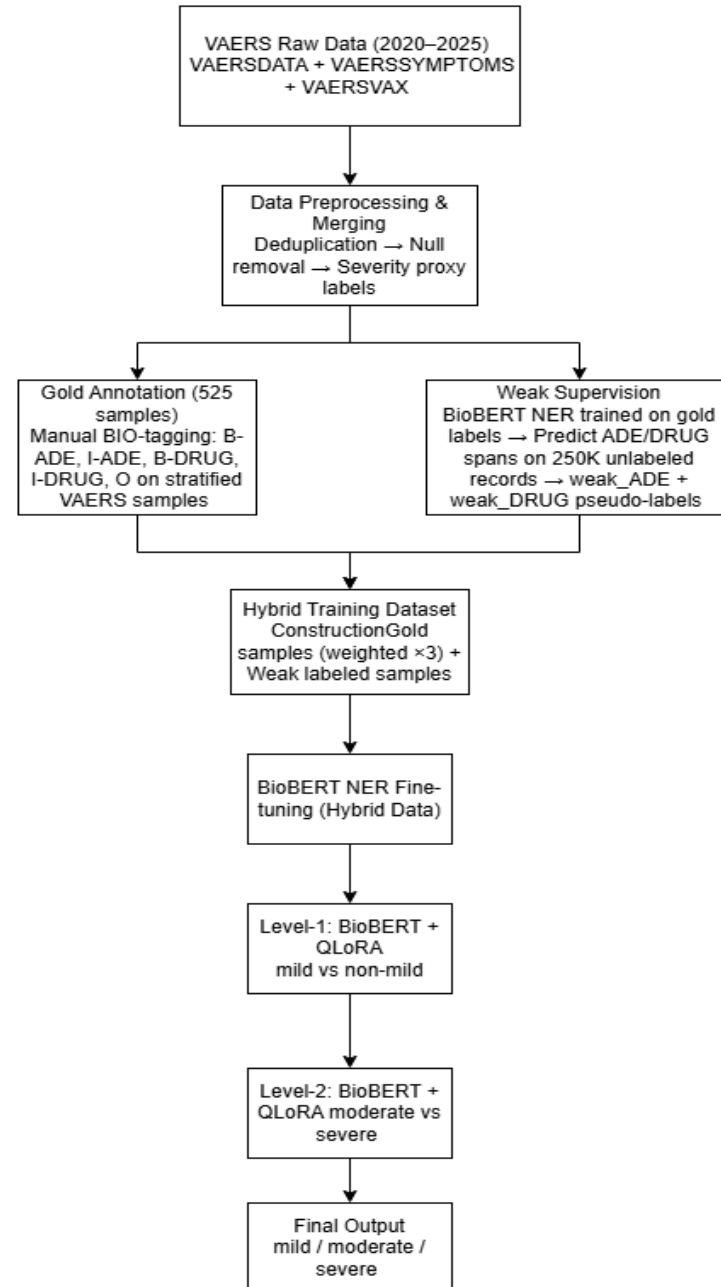
Objective

- To build an end-to-end NLP pipeline that automatically extracts Adverse Drug Event and Drug entities from VAERS reports using BioBERT-based NER with weak supervision, and classifies report severity into mild, moderate, or severe using a two-stage hierarchical classifier fine-tuned with QLoRA.

Dataset Description

- Source: VAERS (Vaccine Adverse Event Reporting System) — publicly available from CDC/FDA.
- Coverage: 2020–2025, three files per year — VAERSDATA, VAERSSYMPTOMS, VAERSVAX — merged on VAERS_ID.
- Scale: 200K+ records after preprocessing — free-text SYMPTOM_TEXT field is the primary NLP input.
- Annotation: Small gold dataset manually annotated in BIO-tag format (B-ADE, I-ADE, B-DRUG, I-DRUG, O); 250K records weakly labeled using the trained NER model.

Methodology



Results(NER Performance Table)

Entity	Precision	Recall	F1-Score
ADE	0.46	0.42	0.44
Drug	0.69	0.91	0.79

Level-1: mild vs non-mild

Class	Precision	Recall	F1-Score
Mild	0.84	0.85	0.84
Non-mild	0.85	0.84	0.84

Level-2: moderate vs severe

Class	Precision	Recall	F1-Score
Moderate	0.81	0.87	0.84
Severe	0.79	0.70	0.74

Thank You